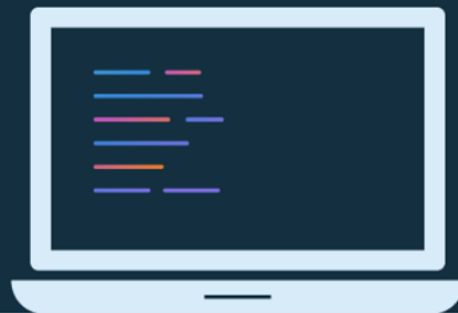


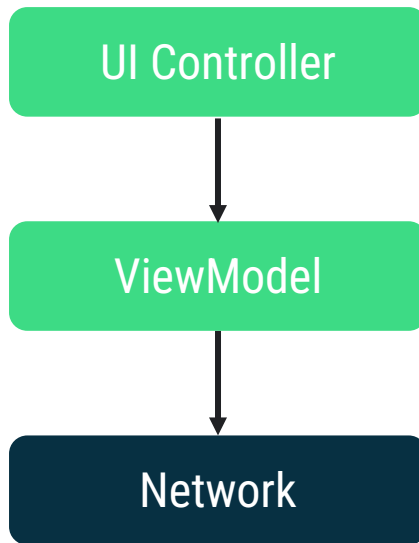


Android Development with Kotlin



Repository pattern

Existing app architecture



Relative data speeds

Operation	Relative speed
Reading from <code>LiveData</code>	FAST
Reading from <code>Room</code> database	SLOW
Reading from network	SLOWEST

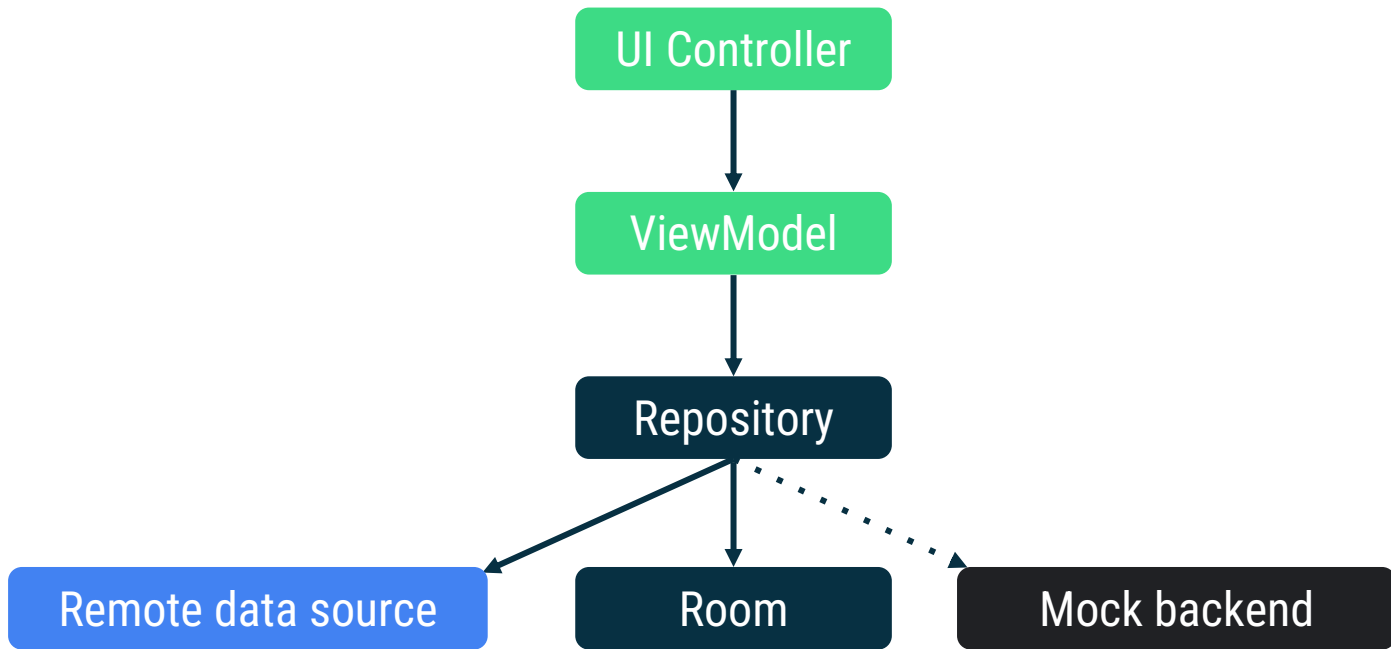
Cache network responses

- Account for long network request times and still be responsive to the user
- Use `Room` locally when retrieval from the network may be costly or difficult
- Can cache based on the least recently used (LRU) value, frequently accessed (FRU) values, or other algorithm

Repository pattern

- Can abstract away multiple data sources from the caller
- Supports fast retrieval using local database while sending network request for data refresh (which can take longer)
- Can test data sources separately from other aspects of your app

App architecture with repository pattern



Implement a repository class

- Provide a common interface to access data:
 - Expose functions to query and modify the underlying data
- Depending on your data sources, the repository can:
 - Hold a reference to the DAO, if your data is in a database
 - Make network requests if you connect to a web service

Demo

Assignment

Apply Repository pattern and full cycle with MVVM for the previous task (task day 8)